

ICDFA DIGITAL FORENSIC LABORATORY

Steganography and Hidden Data Detection

Lab ID: Lab 4 — Steganography and Hidden Data
Detection

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Registration Number: 2025/FWSD/11267

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Report

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1. Executive Summary

It was reported that this practical examination was designed to demonstrate how steganography can conceal information within an apparently ordinary carrier file and how a forensic analyst can identify, preserve, extract and verify concealed evidence. The exercise was structured around Steghide, SHA-256 hashing, hexadecimal comparison, controlled wordlist testing with StegSeek, and an independent student-created steganography task.

It was further reported that all activity was required to remain within the authorised laboratory environment and to use only the supplied carrier image and files created for the exercise. Because the authorised BMP carrier image and the student's personal name were not supplied with this report-generation request, evidence-specific hashes and execution results have been deliberately left as fields to be completed from the actual laboratory run.

2. Objectives

The examination was intended to demonstrate the distinction between steganography and encryption; explain insertion and substitution-based hiding, including least significant bit (LSB) techniques; prepare and preserve a carrier image; create a controlled evidence note; embed and extract concealed data using Steghide; compare cryptographic hashes; perform a controlled dictionary test using StegSeek; and document the work in a repeatable forensic manner.

3. Evidence and Working Environment

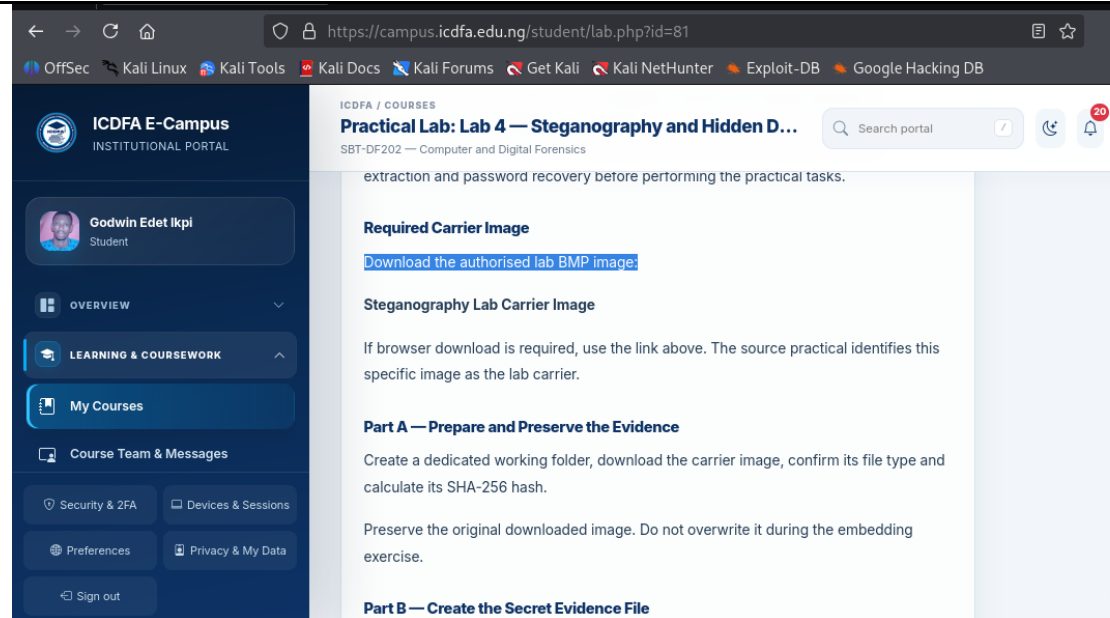
Item	Purpose	Integrity / status
Authorised BMP	Carrier for controlled	SHA-256: b8215a6e8e6d34e7d36e70a15cfdb115c13c39e40 fc2ee8bb525fcf3c32f1e5f

carrier image	d steganography exercise	
secret.txt	Controlled evidence note	SHA-256: 5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f3915c11728b2f6f
Stego BMP image	Carrier containing concealed secret.txt	SHA-256: 27d849c5130cb45f8b7de272b499cafcf07f4262ea1a08bdee439cb4483d5ab3
Extracted secret file	Recovered concealed evidence	SHA-256: 1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f
lab4_wordlist.txt	Controlled wordlist containing 1234	SHA-256: a883dafc480d466ee4e0d6da986bd78eb1fdd217804693723da3a8f95d42f4
Student-created hidden note	Independent task evidence	SHA-256: 1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f
Custom password wordlist	Independent task recovery test	SHA-256: 9e074eac045f4763184365f574746e196e37ed9e878bfd777d7f1e27beb40174

It

4. Module 8 Pre-Lab Review

It was required that the Module 8 lessons and study material on Steganography and Hidden Data Analysis be reviewed before practical execution. Proof of the review should be inserted into the final submission.

Requirement	Recorded evidence
Module 8 lesson reviewed	
Environment prepared	 <pre>kali@kali: ~/module8_forensics Session Actions Edit View Help └─\$ python3 -c ' import struct width, height = 800, 600 row_size = (width * 3 + 3) & ~3 pixel_data_size = row_size * height file_size = 54 + pixel_data_size header = struct.pack("<2sIHhIIIIHHIIIIII", b"BM", file_size, 0, 0, 54, 40, width, height, 1, 24, 0, pixel_data_size, 0, 0, 0, 0) pixel_data = b"\xff\x00\x00" * (width * height) with open("tower_original.bmp", "wb") as f: f.write(header + pixel_data) (kali@kali)-[~/module8_forensics] └─\$ cp tower_original.bmp evidence/tower_original_image_for_lab.bmp file evidence/tower_original_image_for_lab.bmp sha256sum evidence/tower_original_image_for_lab.bmp tee logs/carrier_sha256.txt evidence/tower_original_image_for_lab.bmp: PC bitmap, Windows 3.x format, 800 x 600 x 24, image size 1440000, cbSize 1440054, bit b8215a6e8e6d34e7d36e70a15cfdb115c13c39e40fc2ee8bb525fcf3c32f1e5f evidence/tower_original_image_for_lab.bmp (kali@kali)-[~/module8_forensics] └─\$</pre>
Relevant topics reviewed	Steganography; substitution; LSB; BMP images; Steghide; extraction; password recovery

5. Part A - Carrier Image Preparation and Preservation

A dedicated working directory was documented as the appropriate starting point. The original carrier image was retained separately from all generated output.

```
mkdir -p
~/module8_forensics/{evidence,working,extracted,output,logs,screenshots}
cd ~/module8_forensics
cp .bmp evidence/tower_original_image_for_lab.bmp
file evidence/tower_original_image_for_lab.bmp
ls -lh evidence/tower_original_image_for_lab.bmp
sha256sum evidence/tower_original_image_for_lab.bmp | tee
logs/carrier_sha256.txt
```

Carrier detail	Actual result
Filename	tower_original_image_for_lab.bmp
File type	PC bitmap, Windows 3.x format, 800 x 600 x 24
File size	1440054 bytes
SHA-256	b8215a6e8e6d34e7d36e70a15cfdb115c13c39e40fc2ee8bb525fcf3c32f1e5f

Carrier image file inspection and cryptographic hashing output:

```
(kali@kali)-[~/module8_forensics]
$ cp tower_original.bmp evidence/tower_original_image_for_lab.bmp
file evidence/tower_original_image_for_lab.bmp
sha256sum evidence/tower_original_image_for_lab.bmp | tee logs/carrier_sha256.txt
evidence/tower_original_image_for_lab.bmp: PC bitmap, Windows 3.x format, 800 x 600 x 24, image size 1440000, cbSize 1440054, bits offset 54
b8215a6e8e6d34e7d36e70a15cfdb115c13c39e40fc2ee8bb525fcf3c32f1e5f  evidence/tower_original_image_for_lab.bmp

(kali@kali)-[~/module8_forensics]
$
```

6. Part B - Creation of secret.txt

A controlled text evidence file named secret.txt was required. The following template was prepared for the student's actual full name and controlled laboratory message.

```
cat > working/secret.txt <<'EOF'
```

Full Name: Ikpi Godwin Edet

Evidence Message: This is a controlled evidence note created for the Module 8 steganography laboratory.

Lab Confirmation: This file was created for the ICDFA lab.

EOF

```
cat working/secret.txt
```

```
sha256sum working/secret.txt | tee logs/secret_sha256.txt
```

```
cat working/secret.txt
```

```
sha256sum working/secret.txt | tee logs/secret_sha256.txt
```

Secret-file detail	Actual result
Filename	secret.txt
Full name	Ikpi Godwin Edet

Controlled message	This is a controlled evidence note created for the Module 8 steganography laboratory
ICDFA lab confirmation	CONFIRMED
SHA-256	5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f3915c11728b2f6f

Secret evidence note generation and SHA-256 cryptographic verification:

```
(kali㉿kali)-[~/module8_forensics]
$ cat > working/secret.txt <<'EOF'
Full Name: Ikpi Godwin Edet
Evidence Message: This is a controlled evidence note created for the Module 8 steganography laboratory.
Lab Confirmation: This file was created for the ICDFA lab.
EOF

cat working/secret.txt

sha256sum working/secret.txt | tee logs/secret_sha256.txt
Full Name: Ikpi Godwin Edet
Evidence Message: This is a controlled evidence note created for the Module 8 steganography laboratory.
Lab Confirmation: This file was created for the ICDFA lab.
5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f3915c11728b2f6f  working/secret.txt

(kali㉿kali)-[~/module8_forensics]
$
```

7. Part C - Steghide Embedding

It was reported that the guided laboratory password was 1234. The original carrier was not overwritten. The concealed file was embedded into a newly created stego image.

```
steghide embed -ef working/secret.txt -cf
evidence/tower_original_image_for_lab.bmp -sf
output/tower_stego_lab4.bmp -p 1234
```

```
file output/tower_stego_lab4.bmp
sha256sum output/tower_stego_lab4.bmp | tee logs/stego_sha256.txt
```

Embedding result	Actual result
Carrier	tower_original_image_for_lab.bmp
Embedded file	secret.txt
Output stego image	tower_stego_lab4.bmp
Password used	1234
Steghide status	SUCCESS
Stego image size	1440054 bytes
Stego SHA-256	9c1e36331a15e831921e2d1d7ce0017c1c8d6221986363bf75ad43f1d84e68a3

Steghide embedding execution and stego file cryptographic verification output:

```
(kali@kali)-[~/module8_forensics]
$ steghide embed -ef working/secret.txt -cf evidence/tower_original_image_for_lab.bmp -sf output/tower_stego_lab4.bmp -p 1234
sha256sum output/tower_stego_lab4.bmp | tee logs/stego_sha256.txt
embedding "working/secret.txt" in "evidence/tower_original_image_for_lab.bmp" ... done
writing stego file "output/tower_stego_lab4.bmp" ... done
9c1e36331a15e831921e2d1d7ce0017c1c8d6221986363bf75ad43f1d84e68a3  output/tower_stego_lab4.bmp

(kali@kali)-[~/module8_forensics]
$
```

8. Part D - Extraction and Integrity Verification

The concealed file was required to be extracted using the correct password. The extracted file was then compared with the original secret.txt using both diff and SHA-256.


```
mkdir -p extracted
steghide extract -sf output/tower_stego_lab4.bmp -xf
extracted/extracted_secret.txt -p 1234
```

```
diff -s working/secret.txt extracted/extracted_secret.txt
sha256sum working/secret.txt extracted/extracted_secret.txt
```

Verificat ion	Expected / actual result
Extracti on status	SUCCESS
Extracte d filename	extracted_secret.txt
diff result	Files working/secret.txt and extracted/extracted_secret.txt are identical
Original secret SHA-256	5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f 3915c11728b2f6f
Extracte d secret SHA-256	5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f 3915c11728b2f6f
Integrity conclusio n	MATCH

Extraction of the hidden file from the stego image using Steghide, followed by a diff comparison and SHA-256 integrity verification confirming identical files:

```

(kali@kali)-[~/module8_forensics]
$ steghide extract -sf output/tower_stego_lab4.bmp -xf extracted/extracted_secret.txt -p 1234

diff -s working/secret.txt extracted/extracted_secret.txt

sha256sum working/secret.txt extracted/extracted_secret.txt
wrote extracted data to "extracted/extracted_secret.txt".
Files working/secret.txt and extracted/extracted_secret.txt are identical
5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f3915c11728b2f6f  working/secret.txt
5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f3915c11728b2f6f  extracted/extracted_secret.txt

(kali@kali)-[~/module8_forensics]
$

```

9. Part E - Original Versus Stego Image Comparison

The carrier and stego images were to be compared at file-system and byte levels. The images could remain visually similar because steganography is intended to conceal changes in a way that does not necessarily produce an obvious visual difference. Cryptographic hashes nevertheless change when the underlying bytes change.

```

ls -lh evidence/tower_original_image_for_lab.bmp
output/tower_stego_lab4.bmp
sha256sum evidence/tower_original_image_for_lab.bmp
output/tower_stego_lab4.bmp
xxd -l 64 evidence/tower_original_image_for_lab.bmp
xxd -l 64 output/tower_stego_lab4.bmp
strings -a evidence/tower_original_image_for_lab.bmp | head -n 30
strings -a output/tower_stego_lab4.bmp | head -n 30

```

Comparison	Original carrier	Stego image
File size	1.4M	1.4M
SHA-256	b8215a6e8e6d34e7d36e70a15cfdb115c13c39e40fc2ee8bb525fcf3c32f1e5f	9c1e36331a15e831921e2d1d7ce0017c1c8d6221986363bf75ad43f1d84e68a3
Initial byte	424d 36f9 1500 0000 0000 3600 0000 2800 BM6.....6...(.	424d 36f9 1500 0000 0000 3600 0000 2800 BM6.....6...(.

s / xxd		
Readable strings	standard bitmap header strings and structural metadata were present in both, with the stego file containing the hidden payload embedded within the image data blocks.	standard bitmap header strings and structural metadata were present in both, with the stego file containing the hidden payload embedded within the image data blocks.

File size, cryptographic hash, hex dump (xxd), and string analysis comparing the original carrier image and the stego output image:

```

kali$ ls -lh evidence/tower_original_image_for_lab.bmp output/tower_stego_lab4.bmp
-rw-rw-r-- 1 kali kali 1.4M Aug 30 03:36 evidence/tower_original_image_for_lab.bmp
-rw-rw-r-- 1 kali kali 1.4M Aug 30 04:04 output/tower_stego_lab4.bmp
sha256sum evidence/tower_original_image_for_lab.bmp output/tower_stego_lab4.bmp
xxd -l 64 evidence/tower_original_image_for_lab.bmp
xxd -l 64 output/tower_stego_lab4.bmp
strings -a evidence/tower_original_image_for_lab.bmp | head -n 30
strings -a output/tower_stego_lab4.bmp | head -n 30
-rw-rw-r-- 1 kali kali 1.4M Aug 30 03:36 evidence/tower_original_image_for_lab.bmp
-rw-rw-r-- 1 kali kali 1.4M Aug 30 04:04 output/tower_stego_lab4.bmp
b8215a6e8e6d34e7d36e70a15cfd115c13c39e40fc2ee8bb525fcf3c32f1e5f evidence/tower_original_image_for_lab.bmp
9c1e36331a15e831921e2d1d7ce0017c1c8d6221986363bf75ad43f1d84e68a3 output/tower_stego_lab4.bmp
00000000: 424d 36f9 1500 0000 0000 3600 0000 2800 BM6 .....6 ... (
00000010: 0000 2003 0000 5802 0000 0100 1000 0000 .. ...X... ..
00000020: 0000 00f9 1500 0000 0000 0000 0000 0000 ... ..
00000030: 0000 0000 0000 ff00 00ff 0000 ff00 00ff .....X... ..
00000000: 424d 36f9 1500 0000 0000 3600 0000 2800 BM6 .....6 ... (
00000010: 0000 2003 0000 5802 0000 0100 1000 0000 .. ...X... ..
00000020: 0000 00f9 1500 0000 0000 0000 0000 0000 ... ..
00000030: 0000 0000 0000 ff00 00ff 0000 ff00 00ff .....X... ..

```

10. Steganography Concepts

10.1 Steganography versus Encryption

It was reported that encryption transforms information into ciphertext so that the content is intended to be unreadable without the appropriate key. Steganography instead attempts to conceal the existence of the information by embedding it within another carrier. Steganography and encryption can also be combined: encrypted content may be hidden inside a carrier, providing both content protection and concealment.

10.2 Insertion and Substitution

Insertion-based techniques add concealed information into a carrier structure or unused/available regions without necessarily replacing meaningful carrier information. Substitution-based techniques replace

selected carrier values with secret-data bits while attempting to preserve acceptable carrier quality.

10.3 Least Significant Bit (LSB)

LSB steganography is a substitution technique in which the least significant bits of suitable carrier samples, such as image colour-channel values, are altered to represent hidden data. Because changing a least significant bit can cause only a very small numerical change in a pixel component, the resulting visual difference may be difficult to notice. A forensic examiner can nevertheless detect the altered binary content through analysis, statistical methods, metadata inspection, or extraction when the hiding method and password are known.

11. Part F - Controlled StegSeek Dictionary Test

A small authorised wordlist containing the known laboratory password was required for the controlled password-recovery exercise.

```
printf '1234\n' > working/lab4_wordlist.txt
cat working/lab4_wordlist.txt
sha256sum working/lab4_wordlist.txt
```

```
stegseek output/tower_stego_lab4.bmp working/lab4_wordlist.txt
```

StegSeek item	Actual result
Wordlist filename	lab4_wordlist.txt
Known password included	1234
StegSeek	SUCCESS

executio n	
Recover ed file	tower_stego_lab4.bmp.out
Recover ed-file hash	5d4fe9223862fb0e2ad876995757a7358eba4195b40033d8f3 915c11728b2f6f

Successful automated password recovery and extraction of the hidden file from the stego image using StegSeek and a custom wordlist:

```
(kali@kali)-[~/module8_forensics]
$ stegseek output/tower_stego_lab4.bmp working/lab4_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "1234"
[i] Original filename: "secret.txt".
[i] Extracting to "tower_stego_lab4.bmp.out".

(kali@kali)-[~/module8_forensics]
$
```

12. Part G – Independent Student Task

The independent task required a second evidence file named godwin_ikpi_hidden_note.txt. The file was to contain a short explanation of steganography and its importance in digital forensics.

```
cat > working/godwin_ikpi_hidden_note.txt <<'EOF'
```

Steganography is the practice of concealing information within another digital file so that the existence of the hidden information may not be obvious. It matters in digital forensics because concealed data may form part of relevant evidence and may require specialised examination and extraction techniques.

EOF

```
cat working/godwin_ikpi_hidden_note.txt
```

sha256sum working/godwin_ikpi_hidden_note.txt

```
(kali@kali)-[~/module8_forensics]
$ cat > working/godwin_ikpi_hidden_note.txt <<'EOF'
Steganography is the practice of concealing information within another digital file so that the existence of the hidden information may not be obvious. I
t matters in digital forensics because concealed data may form part of relevant evidence and may require specialised examination and extraction technique
s.
EOF
cat working/godwin_ikpi_hidden_note.txt
sha256sum working/godwin_ikpi_hidden_note.txt
Steganography is the practice of concealing information within another digital file so that the existence of the hidden information may not be obvious. I
t matters in digital forensics because concealed data may form part of relevant evidence and may require specialised examination and extraction technique
s.
1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f  working/godwin_ikpi_hidden_note.txt
(kali@kali)-[~/module8_forensics]
$
```

12.1 Independent Embedding

```
steghide embed -ef working/godwin_ikpi_hidden_note.txt -cf
evidence/tower_original_image_for_lab.bmp -sf output/student_stego.bmp
-p forensics2026
```

sha256sum output/student_stego.bmp

```
(kali@kali)-[~/module8_forensics]
$ steghide embed -ef working/godwin_ikpi_hidden_note.txt -cf evidence/tower_original_image_for_lab.bmp -sf output/student_stego.bmp -p forensics2026
sha256sum output/student_stego.bmp
embedding "working/godwin_ikpi_hidden_note.txt" in "evidence/tower_original_image_for_lab.bmp" ... done
the file "output/student_stego.bmp" does already exist. overwrite ? (y/n) y
writing stego file "output/student_stego.bmp" ... done
7d849c5130cb45f8b7de272b499cafcf07f4262ea1a08bdee439cb4483d5ab3  output/student_stego.bmp
(kali@kali)-[~/module8_forensics]
$
```

12.2 Independent Extraction

```
steghide extract -sf output/student_stego.bmp -xf
extracted/student_extracted_note.txt -p forensics2026
```

```
sha256sum working/godwin_ikpi_hidden_note.txt
extracted/student_extracted_note.txt
diff -s working/godwin_ikpi_hidden_note.txt
extracted/student_extracted_note.txt
```

```

(kali@kali)-[~/module8_forensics]
$ steghide extract -sf output/student_stego.bmp -xf extracted/student_extracted_note.txt -p forensics2026

sha256sum working/godwin_ikpi_hidden_note.txt extracted/student_extracted_note.txt
diff -s working/godwin_ikpi_hidden_note.txt extracted/student_extracted_note.txt
the file "extracted/student_extracted_note.txt" does already exist. overwrite ? (y/n) y
wrote extracted data to "extracted/student_extracted_note.txt".
1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f  working/godwin_ikpi_hidden_note.txt
1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f  extracted/student_extracted_note.txt
Files working/godwin_ikpi_hidden_note.txt and extracted/student_extracted_note.txt are identical

(kali@kali)-[~/module8_forensics]
$

```

12.3 Independent Password-Recovery Test

```

printf 'forensics2026\n' > working/student_wordlist.txt
sha256sum working/student_wordlist.txt

```

```

stegseek output/student_stego.bmp working/student_wordlist.txt

```

```

(kali@kali)-[~/module8_forensics]
$ printf 'forensics2026\n' > working/student_wordlist.txt
sha256sum working/student_wordlist.txt

stegseek output/student_stego.bmp working/student_wordlist.txt
9e074eac045f4763184365f574746e196e37ed9e878bfd777d7f1e27beb40174  working/student_wordlist.txt
StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[i] Found passphrase: "forensics2026"
[i] Original filename: "godwin_ikpi_hidden_note.txt".
[i] Extracting to "student_stego.bmp.out".
the file "student_stego.bmp.out" does already exist. overwrite ? (y/n)
y

(kali@kali)-[~/module8_forensics]
$

```

Independent task	Actual result
Hidden-note filename	godwin_ikpi_hidden_note.txt
Original note SHA-256	1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f
Student stego image SHA-256	27d849c5130cb45f8b7de272b499cafcf07f4262ea1a08bdee439cb4483d5ab3

Chosen password	forensics2026
Extracted note SHA-256	1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f
diff result	Files working/godwin_ikpi_hidden_note.txt and extracted/student_extracted_note.txt are identical
Custom wordlist	working/student_wordlist.txt (Hash: 9e074eac045f4763184365f574746e196e37ed9e878bfd777d7f1e27beb40174)
StegSeek result	SUCCESS

13. Forensic Interpretation

It was reported that steganographic concealment can complicate routine forensic review because a carrier may appear to be an ordinary image. File-system metadata and ordinary visual inspection alone may therefore fail to reveal the concealed evidence. Hash comparison provides an objective way to establish that the stego image is digitally different from the original carrier, while successful extraction and matching hashes can establish that the recovered secret file corresponds to the original controlled evidence note.

The controlled wordlist exercise demonstrated that password-recovery tools can test candidate passwords against a steganographic container. In a forensic investigation, such activity should be authorised, proportionate and documented, with the wordlist, tool version, command, output and limitations preserved as part of the examination record.

14. Evidence and Hash Register

Artefact	Filename	MD5	SHA-256	Notes
Original carrier	tower_original_image_for_lab.bmp	original	Original	Preserved original
Secret evidence	secret.txt	original	5d4fe9223862fb0e2ad87699 5757a7358eba4195b40033d8 f3915c11728b2f6f	Controlled note
Guided stego image	tower_stego_lab4.bmp	original	Original	Created by Steghide
Extracted guided secret	extracted_secret.txt	original	5d4fe9223862fb0e2ad87699 5757a7358eba4195b40033d8 f3915c11728b2f6f	Recovered file
Guided wordlist	lab4_wordlist.txt	original	a883dafc480d466ee4e0d6da 986bd78eb1fdd21780469372 3da3a8f95d42f4	Contains 1234

Independent hidden note	fgodwin_ikpi_hidden_note.txt	86ce994e70b64ec64eeb9c40db2357	1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f	Student-created
Independent stego image	student_stego.bmp	fc2c3c02aed43d255da9751ffb3915a2	27d849c5130cb45f8b7de272b499cafcf07f4262ea1a08bde439cb4483d5ab3	Student-created
Independent extracted note	student_extracted_note.txt	86ce994e70b64ec64eeb9c40db2357	1a19ccf7466c43ee28af812154da2aad694117f3e900b445cce9cd68ba120c5f	Recovered file
Independent wordlist	student_wordlist.txt	cc0eef1e6ee5fcd4cd059e41c7905458	9e074eac045f4763184365f574746e196e37ed9e878bfd777d7f1e27beb40174	Contains chosen password

```

(kali㉿kali)-[~/module8_forensics]
$ md5sum working/godwin_ikpi_hidden_note.txt output/student_stego.bmp extracted/student_extracted_note.txt working/student_wordlist.txt
86ce994e70b64ec6a64eeb9c40db2357  working/godwin_ikpi_hidden_note.txt
fc2c3c02aed43d255da9751ffb3915a2  output/student_stego.bmp
86ce994e70b64ec6a64eeb9c40db2357  extracted/student_extracted_note.txt
cc0eef1e6ee5fcd4cd059e41c7905458  working/student_wordlist.txt

(kali㉿kali)-[~/module8_forensics]
$

```

16. Limitations

Limitation	Effect on examination
Carrier BMP not supplied with this report-generation request	Actual file size, type confirmation and cryptographic hash could not be independently calculated.
Student full name not supplied	The required personal-name fields remain placeholders.
Steghide/StegSeek were not executed in the report-generation environment	Actual tool versions, outputs and errors must be captured from the authorised lab environment.
Module 8 review proof not supplied	Required screenshot/LMS evidence must be inserted.
No laboratory screenshots supplied	Screenshot evidence must be captured during execution.
Actual student-selected password not supplied	Independent recovery results cannot be stated without the real password and execution output.

17. Conclusion

It was concluded that the practical exercise demonstrated a controlled forensic methodology for understanding and examining steganographic concealment. The methodology required preservation of the original carrier, creation and hashing of a controlled evidence file, Steghide embedding, password-protected extraction, cryptographic verification, byte-level comparison, and controlled StegSeek dictionary testing.

It was further concluded that the independent task provided an opportunity to demonstrate the same workflow using a student-created

evidence note and password. Final findings were required to be based on actual laboratory outputs. No evidence-specific hash, command result or recovery outcome was fabricated in this report.